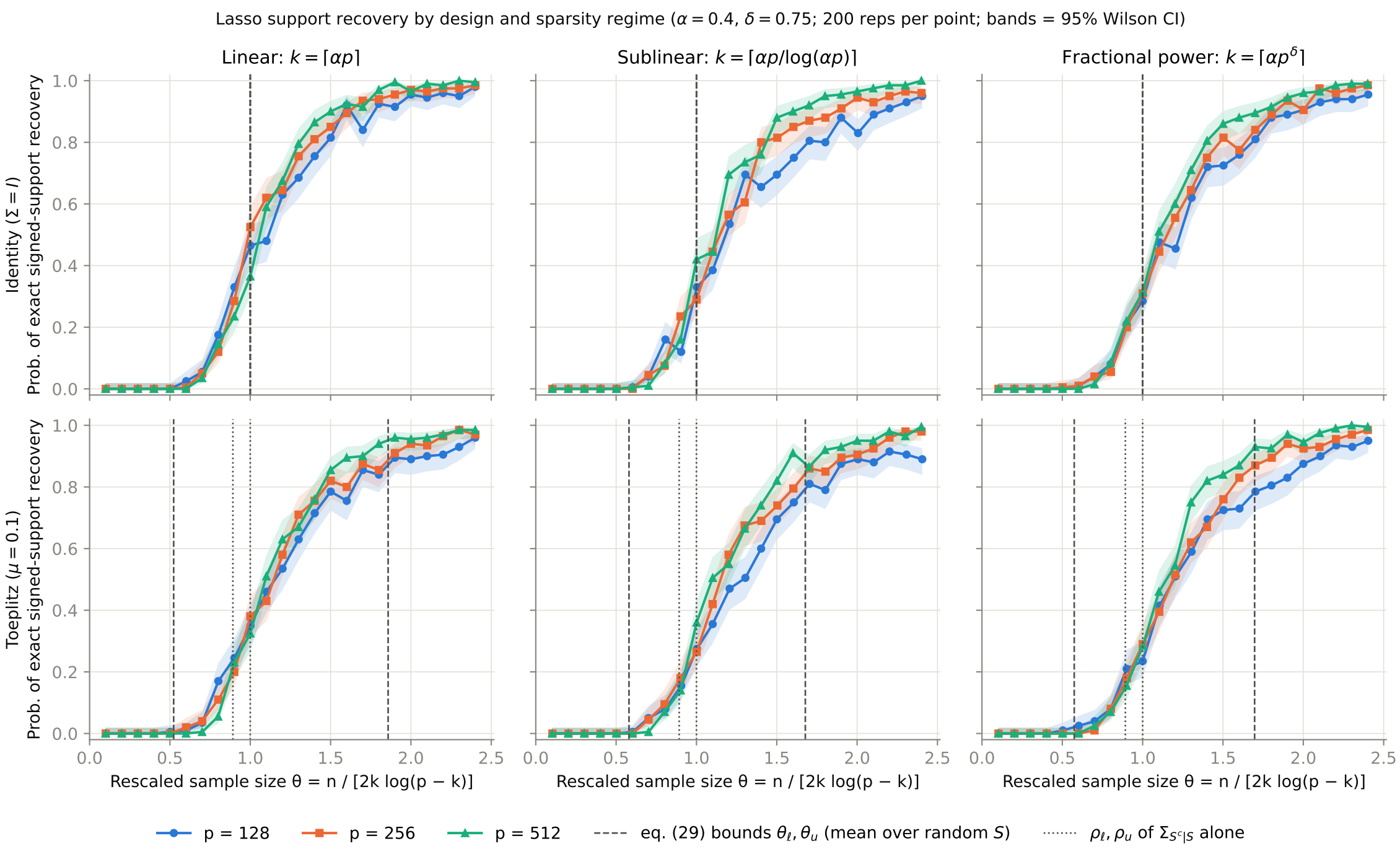
Lasso support recovery by design and sparsity regime ($\alpha = 0.4$, $\delta = 0.75$; 200 reps per point; bands = 95% Wilson CI)
Linear: $k = \lceil \alpha p \rceil$
Sublinear: $k = \lceil \alpha p / \log(\alpha p) \rceil$
Fractional power: $k = \lceil \alpha p^{\delta} \rceil$
Identity ($\Sigma = I$)
Toeplitz ($\mu = 0.1$)
Prob. of exact signed-support recovery
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
Rescaled sample size θ = n / [2k log(p − k)]
p = 128
p = 256
p = 512
eq. (29) bounds θ_ℓ, θ_u (mean over random S)
ρ_ℓ, ρ_u of $\Sigma_{S^c|S}$ alone